

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: Observation Task-01

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1. TITLE

Design and Develop a Webpage using HTML5 and CSS3

2. AIM

To develop web applications utilizing HTML5 semantic tags, forms, multimedia, tables, and Canvas, and to style them using CSS3 properties.

3. OBJECTIVE

- To understand and implement HTML5 semantic elements.
- To create structured tables, interactive forms, and embed audio/video.
- To utilize HTML5 Canvas and JavaScript for basic drawing.
- To apply CSS3 for styling, including flexbox, gradients, box-shadow, and hover effects.

4. THEORY

HTML5 introduces semantic elements (like <header>, <nav>, <section>, <article>, <footer>) that clearly describe their meaning to both the browser and the developer. It supports native multimedia via <audio> and <video> tags, and provides the <canvas> element for drawing graphics using JavaScript.

CSS3 brings advanced styling capabilities to web pages. Features such as rounded corners (border-radius), drop shadows (box-shadow), gradients, and flexible layouts (Flexbox) allow developers to create visually appealing, modern, and responsive user interfaces.

5. ALGORITHM / PROCEDURE

1. Create a project folder containing sub-folders for Task 1 and Task 2.

2. For Task 1, create an index.html file structured with HTML5 semantic tags.
3. Implement the College Information Portal by adding tables, lists, and a comprehensive registration form with various HTML5 input types.
4. Embed multimedia files using <audio> and <video> tags.
5. Apply external CSS styling via style.css to format the layout and elements of the portal.
6. For Task 2, create a new HTML document featuring an HTML5 <canvas> element and styled content cards.
7. Write JavaScript code to draw primitives (rectangle, circle, line, text) onto the canvas.
8. Use CSS3 to style the content cards with gradients, borders, and hover animations.
9. Execute the HTML files in a web browser, test the functionalities, and verify the outputs.

6. PROGRAM

index.html

HTML

```
<!DOCTYPE html>
<!-- TASK 1: College Information Portal -->
<!-- Demonstrates: HTML5 Structure, Text Formatting, Navigation, and Semantic Elements -->
<html lang="en">
<head>
  <!-- Character encoding for proper text rendering -->
  <meta charset="UTF-8">
  <!-- Viewport configuration for responsive layout on mobile devices -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
  <!-- External CSS Stylesheet connection -->
  <link rel="stylesheet" href="../style.css">
</head>
<body>

  <!-- Header Section: Main website banner and branding -->
  <header>
    <h1>Welcome to ABC College of Engineering</h1>
    <p>College Information Portal</p>
  </header>

  <!-- Navigation Bar: Accessible link menu for main sections -->
  <nav>
    <ul class="nav-list">
      <li><a href="#about">About</a></li>
      <li><a href="#students">Students</a></li>
      <li><a href="#registration">Registration</a></li>
      <li><a href="#media">Media</a></li>
    </ul>
  </nav>

  <!-- Main Content Area: Encloses primary page content -->
  <main>
    <!-- About Section -->
    <section id="about">
      <h2>Part A: About the College</h2>
```

```

        <!-- Article container for self-contained institutional details -->
        <article>
            <h3>College Information</h3>
            <p>ABC College of Engineering was established in <b>1998</b>...</p>
        </article>
    </section>
</main>

<!-- Footer Section: Copyright and metadata -->
<footer>
    <p>&copy; 2026 ABC College of Engineering. All Rights Reserved.</p>
</footer>

</body>
</html>

```

style.css

```

CSS

/* ===== Task 1: style.css ===== */

/* Global Body Styles: Sets basic resets, default font, background, and text colors */
body {
    font-family: Arial, Helvetica, sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f4f6f8; /* Light neutral gray background */
    color: #222;               /* Off-black text color for optimal contrast */
}

/* Header Styles: Dark blue background with centered white text */
header {
    background-color: #002855;
    color: #fff;
    text-align: center;
    padding: 20px;
}

/* Navigation Container: Secondary blue shade */
nav {
    background-color: #003f7d;
}

/* Navigation List: Horizontal layout via Flexbox with centered items */
.nav-list {
    list-style: none; /* Remove default bullet points */
    display: flex;
    justify-content: center;
    margin: 0;
    padding: 10px 0;
}

/* Navigation Item Spacing */
.nav-list li {
    margin: 0 15px;
}

/* Navigation Link Formatting: Bold white links without underlines */
.nav-list a {
    color: #fff;
    text-decoration: none;
    font-weight: bold;
}

```

Task 2: HTML5 Canvas & Web Styling

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Styled Webpage</title>
  <!-- Link to external stylesheet for layout styling -->
  <link rel="stylesheet" href="./style.css">
</head>
<body>

  <!-- Header Banner -->
  <header>
    <h1>My Styled Webpage</h1>
    <p>HTML5 Graphics and CSS3 Web Styling</p>
  </header>

  <!-- Section for HTML5 Canvas API Rendering -->
  <section class="canvas-section">
    <h2>Part A: HTML5 Canvas Drawing</h2>
    <!-- Canvas Element: Acts as a drawing surface via JavaScript -->
    <canvas id="myCanvas" width="500" height="300"></canvas>
  </section>

  <!-- Section for UI Components / Card Layouts -->
  <section class="cards-container">
    <div class="card">
      <h2>Card One</h2>
      <p>This card demonstrates border radius, box shadow, and hover effects using CSS3.</p>
      <button>Learn More</button>
    </div>
  </section>

  <!-- Footer Section -->
  <footer>
    &copy; 2026 My Styled Webpage | Created by Vijay Anudeep
  </footer>

  <!-- Inline JavaScript: HTML5 Canvas Drawing Logic -->
  <script>
    // Retrieve canvas element reference and initialize 2D rendering context
    const canvas = document.getElementById('myCanvas');
    const ctx = canvas.getContext('2d');

    // 1. Draw a filled blue rectangle (x: 30, y: 30, width: 150, height: 80)
    ctx.fillStyle = '#4b6cb7';
    ctx.fillRect(30, 30, 150, 80);

    // 2. Draw a filled yellow circle (centerX: 300, centerY: 70, radius: 50)
    ctx.beginPath();
    ctx.arc(300, 70, 50, 0, Math.PI * 2);
    ctx.fillStyle = '#ffe600';
    ctx.fill();

    // 3. Render styled text onto the canvas (x: 130, y: 250)
    ctx.font = "bold 30px 'Georgia', serif";
    ctx.fillStyle = '#182848';
    ctx.fillText("Vijay Anudeep", 130, 250);
  </script>
```

```
</body>
</html>
```

7. CODE EXPLANATION

Code / Tag	Explanation
<header>, <nav>, <footer>	HTML5 semantic elements used to structure the layout.
<audio>, <video>	Used to embed sound and video content natively.
<canvas>	HTML5 element used to draw graphics via JavaScript.
ctx.fillRect()	JavaScript Canvas API to draw a filled rectangle.
display: flex	CSS property to enable a flex container for responsive layout.
box-shadow	CSS3 property to add shadow effects to elements (e.g., cards).
linear-gradient()	CSS function to create an image representing a linear transition between colors.

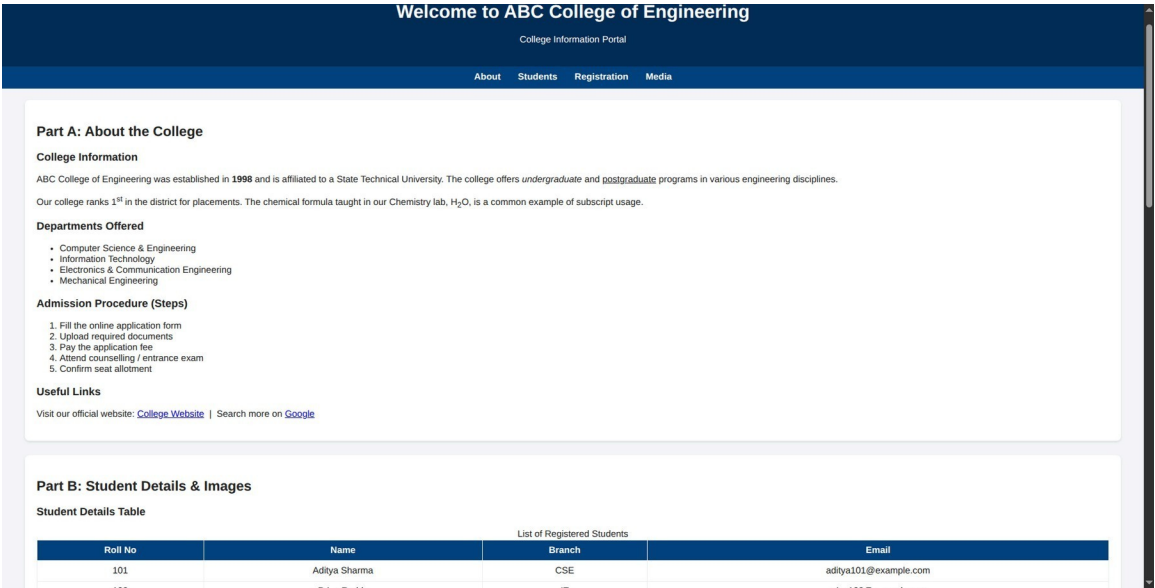
8. TEST CASES AND EXECUTION DATA

Test Case ID	Action / Input	Expected Result	Status
TC-01	Submit Registration Form with all valid inputs.	Form validates required fields and submits successfully.	Pass
TC-02	Leave required fields (e.g., Name, Email) blank and click Submit.	HTML5 form validation triggers; submission is blocked with an error prompt.	Pass
TC-03	Enter an invalid email string (e.g., user@domain) in the email input and submit.	Native HTML5 browser validation alerts user to enter a valid email address.	Pass
TC-04	Click Play on the embedded <audio> and <video> players.	Audio and video media files play smoothly without playback errors.	Pass
TC-05	Attempt to play multimedia with an incorrect or missing file path (src="invalid.mp4").	Browser displays standard fallback controls or error indicators (e.g., media unplayable).	Fail
TC-06	Load page and inspect <canvas id="myCanvas">.	Rectangle, circle, and text "Vijay Anudeep" render accurately at assigned coordinates.	Pass
TC-07	Disable JavaScript or introduce a syntax error in the script block.	Canvas remains blank (300x150 fallback container size) without drawing any shapes.	Fail

Test Case ID	Action / Input	Expected Result	Status
TC-08	Move cursor over .card element.	Card elevates with CSS smooth transition and elevated box-shadow.	Pass
TC-09	Click on #about, #students, #registration, or #media nav links.	Browser smoothly jumps or scrolls to the corresponding section id.	Pass
TC-10	Unlink or rename style.css so the link tag fails to load.	Page renders unstyled semantic raw HTML content without navigation flex layout or header colors.	Fail
TC-11	Resize window below 480px width or view on mobile viewport.	Content wraps within screen bounds without horizontal scrollbar overflow.	Pass

9. OUTPUT

Output 1: College Information Portal



Output 2: My Styled Webpage (Canvas and Cards)

My Styled Webpage

HTML5 Graphics and CSS3 Web Styling

[Home](#)[About](#)[Services](#)[Contact](#)

Part A: HTML5 Canvas Drawing



Vijay Anudeep

Card One

This card demonstrates border radius, box shadow, and hover effects using CSS3.

[Learn More](#)

Card Two

Gradient backgrounds and text shadows are applied to enhance visual appeal.

[Learn More](#)

Card Three

Margins, padding, and fonts are combined to create a clean card layout.

[Learn More](#)

10. RESULT

Thus, web pages demonstrating HTML5 features (semantic tags, forms, multimedia, canvas) and styled using CSS3 were successfully developed. The interactive elements were processed, and the application was verified using multiple test cases.